higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

ELECTRICAL SYSTEMS AND CONSTRUCTION
NQF LEVEL 3

11 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1

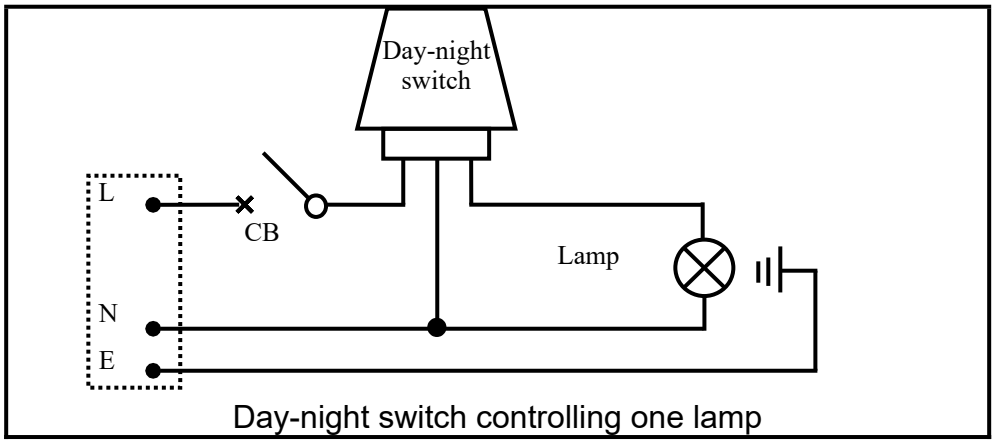
1.1 *Rated current* refers to the current that an electrical device can carry under specified conditions✓ without resulting in overheating or damage to the device.✓ (2)

1.2
- Earthing and bonding
- Electrical separation of circuits
- The use of an isolating transformer with an output of 50 V or less
- The use of electrical equipment that is double insulated (4)

1.3
- It must be installed in the same room as the appliance.
- It must be installed at a height of not less than 0,5 m and not more than 2,2 m above floor level.
- Preferably, it must not be installed above the cooking appliance.
- It must be within 3 m of the appliance, but if the purpose of the switch disconnectors is not indicated clearly, it must be within 0,5 m of the appliance.
- It must not be fixed to the appliance. (Any 3 × 1) (3)

1.4
- The exact paths of conduit and cable runs
- The size and type of conduit to be used
- The size and type of cables to be used
- The positioning of distribution boards, socket outlets, luminaires, geyser supply and stove supply
- Wiring diagrams of the distribution boards. (Any 2 × 1) (2)

1.5 1.5.1



Day-night switch controlling one lamp (6)

1.5.2 10 A (1)

1.6
- Lay-in wireways
- Feed-through wireways
- Wiring trough wireways (3 × 1) (3)

1.7 1.7.1 It is to remove burrs on the inside of a steel conduit.

1.7.2 The conductors may be cut when drawn in.

(2 × 2) (4)

[25]

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QUESTION 2

2.1
- Main circuit breaker
- Earth leakage relay
- Circuit breakers of various sizes
- Surge protection devices
- Load control relay
- Electronic timer (Any 3 × 1) (3)

2.2 2.2.1 The point at which a supplier supplies electricity to any premises (2)

2.2.2 The point at which the consumer can switch the electrical installation on and off from the point of supply. (2)

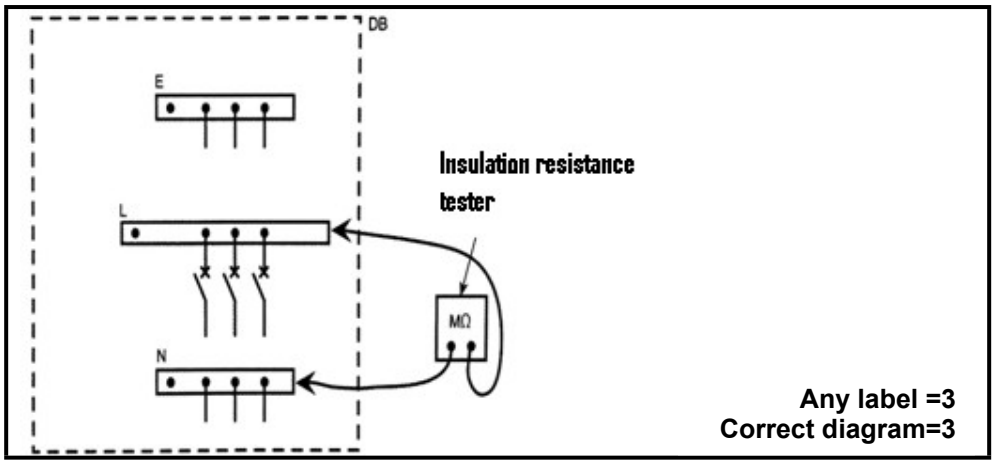
2.3
- Only use test instruments that have been SABS approved.
- Test instruments should be regularly calibrated and maintained.
- Make sure you know the limitations of the respective meters.
- Measuring instruments must be accurate to within 5% or better.
- Make sure you know how to use the respective meters.
- Before using a meter, read the operation manual and keep it on hand if you need to refer to it. (Any 4 × 1) (4)

2.4 2.4.1
- The main should be off.✓
- Measure the voltage between the supply neutral✓ and any earth external to the installation.✓ (3)

2.4.2
- Disconnect the installation
- Inform the supplier. (2 x 1) (2)

2.4.3 The reading should not be more than 25 V. (1)

2.5



Any label =3
Correct diagram=3 (6)

2.6
- The user
- The lessor (Any 1 × 1) (1)

2.7
- A registered electrician
- An accredited electrician (Any 1 × 1) (1)

[25]

QUESTION 3

3.1 Step 1: Do a pre-assessment
Step 2: Observe the equipment
Step 3: Locate the fault
Step 4: Identify the cause of the problem
Step 5: Rectify the fault 5 × 1) (5)

3.2
- Faulty circuit breaker
- Faulty earth leakage relay
- Open-circuit in conductors
- Insulation failure
- Loose wiring
- Damaged socket outlet (Any 3 × 1) (3)

3.3
- Block diagrams
- Schematic or circuit diagrams
- Wiring diagrams (3)

3.4 It is scheduled servicing✓ and inspection of electrical equipment and systems procedures.✓ (2)

3.5
- Planning is an important part in helping to avoid costly mistakes.
- Planning ensures that time and resources are used more efficiently.
- Planning always leads to a purpose, namely the attainment of goals.

(Any 2 × 1) (2)
[15]

QUESTION 4

4.1 4.1.1 False
4.1.2 True
4.1.3 True
4.1.4 False
4.1.5 True

(5 × 1) (5)

4.2
- Bare copper
- Galvanised steel
- Stainless steel (Any 1 × 1) (1)

4.3
- The person the permit is addressed to
- Who will be present throughout the task
- The exact equipment which has been disconnected and its location
- Earthing connections
- The nature of the work to be carried out
- The points of isolation (Any 2 × 1) (2)

4.4 A conductor that functions as both a protective conductor✓ and a neutral conductor.✓ (2)

[10]

QUESTION 5

5.1
5.1.1 E
5.1.2 G
5.1.3 D
5.1.4 A
5.1.5 B

(5 × 1) (5)

5.2 $PermissibleVoltageDrop = 5\% of 230\,V$
$PermissibleVoltageDrop = 11,5\,V$
$PermissibleVoltageDrop = 230 - 11,5$
$PermissibleVoltageDrop = 218,5\,V$ (3)

5.3 If the lamp is switched off, it cannot be restarted immediately as the vapour pressure is too high to allow ionisation✓ and must be allowed to cool down.✓ (2)

5.4
- Performing checks on all connections and wiring
- Cleaning the lighting system
- Checking the lighting load on each circuit
- Updating the lighting system maintenance log sheet (Any 3 × 1) (3)

5.5
- Ripple relays
- Load control relays (2 x 1) (2)

[15]

QUESTION 6

6.1
- Helps identify training needs
- Takes steps to ensure that the worksite is safe to prevent injuries (2)

6.2 It prevents the battery from discharging✓ should there be no sunlight.✓ (2)

6.3
- The input energy is free.
- They require very little maintenance.
- They have no moving parts.
- They are reliable.
- They are easily manageable and upgradeable.
- They can be easily installed at existing installations. (Any 3 × 1) (3)

6.4
- The size and type of the system
- The amount of sun available at the site
- The quality of workmanship during installation
- The tilt angle and orientation of the collectors (Any 3 × 1) (3)

[10]

TOTAL: 100